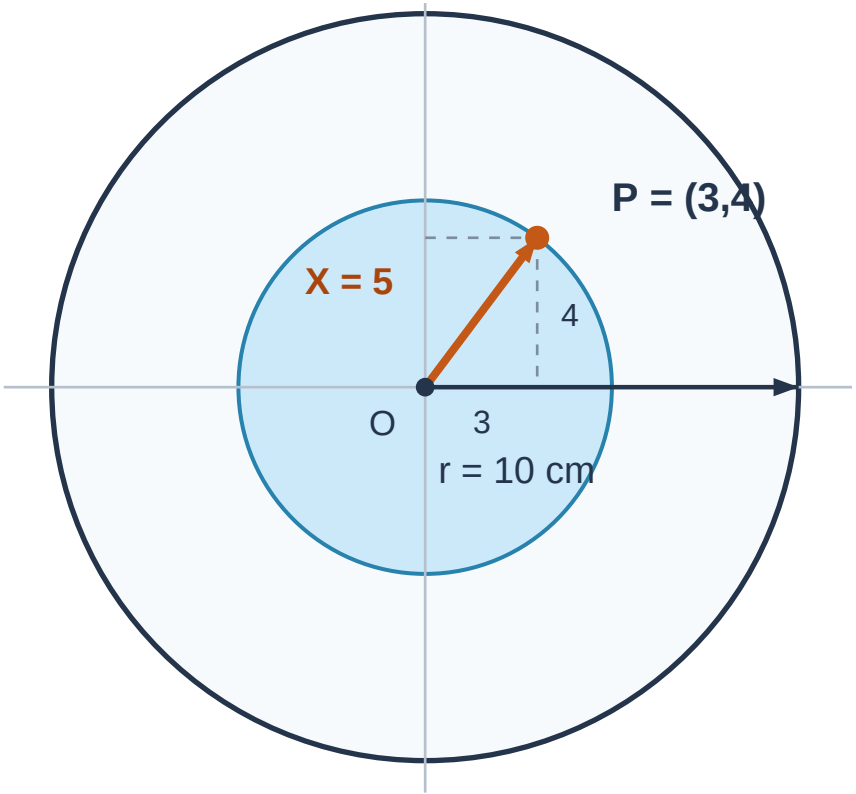

One point, one distance

Outcome: P = (3,4)

The dart lands at this point.

Random variable: X(P) = 5 cm

The distance from P to O.

Blue region: distance at most 5 cm

All points in the inner disc.

This is the preimage of [0,5].